Radar Search Pipeline TO BE 0.7.6.4.18.1.2

Review design for a planned Radar search pipeline change

Status: TO BE

Slice: 0.7.6.4.18.1.2: Live extraction robustness and post-extraction salvage

Pipeline: candidate-discovery

Source of truth for current AS IS: docs/radar/RADAR_SEARCH_PIPELINE_AS_IS.md

Generated PDF: docs/radar/to-be/RADAR_SEARCH_PIPELINE_TO_BE_0.7.6.4.18.1.2.pdf

1. Decision Context

After the recall-first upstream admission slice, deterministic and recorded tests prove that candidate discovery can retain source-backed upstream leads without signal evidence. A bounded Docker/API benchmark smoke exposed an earlier live-provider blocker: OpenRouter can return schema-invalid extraction content while its product-safe source annotations already contain useful candidate evidence. The run then stops before expansion and benchmark target-funnel diagnostics.

This slice merges the older 0.7.6.3.6.6 backlog item into the current candidate-discovery corrective track. It does not implement signal-monitoring runtime.

2. Intended Behavior

Candidate discovery keeps strict extraction validation and bounded retry/backup attempts. If those attempts still leave the extraction contract invalid, the pipeline performs one deterministic post-extraction salvage pass before terminal checkpoint stop.

Salvage may use only product-safe source diagnostics:

- normalized RadarSourceEvidence title, snippet, URL, and source ref;
- retrieved or analyzed source title/snippet/URL metadata;
- source lifecycle diagnostics that already exclude prompts, secrets, headers,

tokens, hidden reasoning, and raw private provider payloads.

If source diagnostics contain an explicit source-backed legal/company lead, candidate discovery materializes a review-needed upstream observation and re-runs the checkpoint. If no safe source-backed lead exists, the run remains stopped for review with an explicit unrecovered reason.

3. Changed Flow

4. Roles Changed

ExtractionFailureClassifier owns provider-neutral failure categories: schema_invalid_empty, schema_invalid_with_sources, unlinked_source_refs, backup_schema_invalid, retry_budget_exhausted, and unrecoverable_no_source_text.

PostExtractionSalvageService owns deterministic salvage from product-safe source diagnostics. It does not call providers and does not accept downstream product candidates.

Checkpoint recovery invokes salvage only at the bounded extraction-recovery limit. It records the result and re-checkpoints recovered state.

5. Diagnostic Surface

Execution metadata adds:

- post_extraction_salvage_records;
- post_extraction_salvage_count;
- post_extraction_salvage_outcome;
- post_extraction_salvage_unrecovered_reason.

Existing extraction validation issues, repair records, model attempts, retry records, and budget decisions remain visible. A recovered run reports extraction_contract_state=post_extraction_salvage_recovered.

6. Semantics And Boundaries

Salvaged candidates are upstream review leads:

- upstream_discovery_outcome=review_needed_upstream_lead;
- product_acceptance_status=review_required;
- no signal search provider calls;
- no not_observed projection;
- no product acceptance without strict qualification evidence.

No source ref or no product-safe source text means no candidate is created. This is not a hidden broad fallback.

7. Test Plan

Fast tests must prove:

- empty schema-invalid output still stops without hidden fallback;
- source-backed schema-invalid output creates review-needed upstream leads;
- unlinked refs and retry/backup exhaustion are separately classified;
- checkpoint recovery continues after salvage instead of terminal schema_failed;
- benchmark smoke can reach expansion and target-funnel diagnostics before a longer live benchmark is attempted.

8. Out Of Scope

- No signal-monitoring live runtime/API work.
- No quality claim from one live run.
- No SIBUR-specific production hardcode outside benchmark fixtures.
- No broad prompt redesign unless a focused test requires a minimal repair

prompt change.